Extend nonuniform public-spirit distortion monotonicity to four or more alternatives

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Abstract

This paper gives a complete solution to the componentwise public-spirit monotonicity problem. For every positive number of voters, every finite number of alternatives, every deterministic voting rule, and every pair of public-spirit vectors satisfying $\gamma' \geq \gamma$ componentwise, we prove

$$\mathrm{dist}(f, \gamma') \leq \mathrm{dist}(f, \gamma).$$

Thus increasing public spirit cannot increase worst-case utilitarian distortion. The proof applies in particular when there are at least four alternatives, and in fact establishes the result for every finite number of alternatives. This is a complete solution, not partial progress; no case with $m \geq 4$ remains unresolved under the stated model.

1 The open problem

This paper addresses an open problem posed by Bailey Flanigan, Ariel D. Procaccia, Sven Wang in *Distortion Under Public-Spirited Voting* [1] (Economics and Computation (EC), 2023), Section 4.2 (Nonuniform PS-monotonicity), page 13 (around Proposition 4.5 discussion).. The website catalogue entry is problem 9 (`W4383533465_p0`) of the Open Problems in Operations Research collection.

> *Fix $m \geq 4$. Determine whether the following monotonicity inequality holds for* all *deterministic voting rules f: for every number of voters n and every pair of PS-vectors $\gamma, \gamma' \in [0,1]^n$ with $\gamma' \geq \gamma$,*
>
> $$\mathrm{dist}(f, \gamma') \leq \mathrm{dist}(f, \gamma).$$
>
> *Equivalently, decide whether increasing voters' public spirit levels componentwise can ever increase the worst-case utilitarian distortion when $m \geq 4$.*

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

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*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.9 write-up; run `run_20260824_032648_50ab97d0, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

2 Model and main theorem

Let $N = [n]$ be a set of $n \geq 1$ voters and let A be a set of $m \geq 1$ alternatives. A utility instance is a matrix

$$U = (u_i(a))_{i \in N,\ a \in A} \in [0,1]^{N \times A}.$$

The utilitarian social welfare of a is

$$S_a = \mathrm{sw}(a, U) := \sum_{i=1}^{n} u_i(a).$$

A public-spirit vector is $\gamma = (\gamma_1, \ldots, \gamma_n) \in [0,1]^n$. Voter i's public-spirit score for alternative a is

$$V_i^{\gamma}(a; U) := (1 - \gamma_i) u_i(a) + \frac{\gamma_i}{n} S_a.$$

Let $\mathcal{R}(A)$ denote the set of strict total orders on A, and let $\pi = (\pi_1, \ldots, \pi_n) \in \mathcal{R}(A)^n$ be a strict ordinal profile. We say that π is compatible with (U, γ) if

$$a \succ_{\pi_i} b \quad \Longrightarrow \quad V_i^{\gamma}(a; U) \geq V_i^{\gamma}(b; U)$$

for every $i \in N$ and every $a, b \in A$. Thus ties in public-spirit scores may be broken adversarially into any compatible strict order. Write $\Pi(U, \gamma)$ for the set of compatible profiles.

A deterministic voting rule is an arbitrary function

$$f : \mathcal{R}(A)^n \longrightarrow A.$$

Its worst-case utilitarian distortion at γ is

$$\mathrm{dist}(f, \gamma) := \sup_{\substack{U \in [0,1]^{N \times A} \\ U \neq 0}} \ \max_{\pi \in \Pi(U, \gamma)} \frac{\max_{a \in A} S_a}{S_{f(\pi)}}.$$

The ratio is interpreted in the extended nonnegative reals: it is $+\infty$ when the numerator is positive and the denominator is zero. Excluding the all-zero utility matrix removes the only possible $0/0$ expression.

Theorem 1 (Complete componentwise monotonicity)**.** *Let $n, m \geq 1$, let $f : \mathcal{R}(A)^n \to A$ be any deterministic voting rule, and let $\gamma, \gamma' \in [0,1]^n$. If*

$$\gamma'_i \geq \gamma_i \qquad \text{for every } i \in N,$$

then

$$\mathrm{dist}(f, \gamma') \leq \mathrm{dist}(f, \gamma).$$

For $m = 1$, the unique alternative is both the winner and a welfare maximizer, so the distortion is identically 1. We therefore assume $m \geq 2$ throughout the substantive proof.

3 Homogeneous and profilewise reduction

All compatibility inequalities are homogeneous in U, and welfare ratios are unchanged when every utility is multiplied by the same positive scalar. Consequently, the restriction $u_i(a) \leq 1$ does not affect the supremum. Indeed, every nonzero finite nonnegative matrix can be multiplied by a sufficiently small positive scalar to place it in $[0,1]^{N\times A}$, without changing its compatible profiles or welfare ratios. We may therefore work on the full nonnegative cone.

Fix a strict profile π and an alternative w. Define

$$D_{\pi,w}(\gamma) := \sup_{\substack{U\geq 0,\ U\neq 0\\ \pi\in\Pi(U,\gamma)}} \frac{\max_{a\in A} S_a}{S_w}.$$

There are only finitely many strict profiles, and every strict profile is feasible: taking all utilities equal and positive makes all public-spirit scores equal. Hence

$$\mathrm{dist}(f,\gamma) = \max_{\pi\in\mathcal{R}(A)^n} D_{\pi,f(\pi)}(\gamma). \tag{1}$$

It is therefore enough to prove

$$D_{\pi,w}(\gamma') \leq D_{\pi,w}(\gamma) \tag{2}$$

for every fixed pair (π, w).

We first prove (2) when all public-spirit coordinates are strictly below 1. For such a vector define

$$r_i = \frac{\gamma_i}{n(1-\gamma_i)} \in [0,\infty). \tag{3}$$

Multiplication of voter i's public-spirit score by the positive constant $1/(1-\gamma_i)$ shows that compatibility with π_i is equivalent to

$$u_i(a) + r_i S_a \geq u_i(b) + r_i S_b \qquad \text{whenever } a \succ_{\pi_i} b. \tag{4}$$

4 A normalized linear program

Fix $t \in A \setminus \{w\}$. Let $q_t(r)$ be the optimum of the linear program

$$\begin{aligned}
\text{maximize}\quad & S_t, && \\
\text{subject to}\quad & u_i(a) + r_i S_a \geq u_i(b) + r_i S_b && \text{if } a \succ_{\pi_i} b,\\
& u_i(a) \geq 0 && \text{for all } i, a,\\
& S_a = \sum_{i=1}^{n} u_i(a) && \text{for all } a,\\
& S_t \geq S_a && \text{for all } a,\\
& S_t + S_w = 1. &&
\end{aligned} \tag{5}$$

The feasible region is nonempty: taking $u_i(a) = 1/(2n)$ for every i, a gives $S_a = 1/2$ for every a. It is compact, because

$$0 \leq S_a \leq S_t \leq 1 \quad \text{and} \quad 0 \leq u_i(a) \leq S_a \leq 1.$$

Thus the optimum is attained. Moreover,

$$\frac{1}{2} \le q_t(r) \le 1. \tag{6}$$

The lower bound follows from the all-equal feasible point, while the upper bound follows from $S_t + S_w = 1$ and nonnegativity.

Lemma 1 (Exact distortion representation)**.** *For $m \ge 2$,*

$$D_{\pi,w}(r) = \max_{t \in A \setminus \{w\}} \frac{q_t(r)}{1 - q_t(r)}, \tag{7}$$

where the right-hand side is $+\infty$ if some $q_t(r) = 1$.

Proof. Consider any nonzero compatible utility matrix and let t be a welfare-maximizing alternative. If $t \neq w$, then $S_t + S_w > 0$, so homogeneity permits the normalization

$$S_t + S_w = 1.$$

Under this normalization,

$$\frac{\max_a S_a}{S_w} = \frac{S_t}{1 - S_t}.$$

The normalized matrix is feasible for (5). Conversely, every feasible point of (5) is a compatible instance in which t is welfare-maximizing. Since $x \mapsto x/(1-x)$ is increasing on $[1/2, 1]$, the largest ratio on the t-maximal face is

$$\frac{q_t(r)}{1 - q_t(r)}.$$

If w itself is welfare-maximizing, the ratio equals 1. By (6), every term on the right-hand side of (7) is at least 1, so this additional case is already covered. If $q_t(r) = 1$, the corresponding optimum has $S_w = 0$ and $S_t = 1$, giving infinite distortion. This proves (7). □

It remains to show that every $q_t(r)$ is componentwise nonincreasing.

5 The dual sign lemma

Only adjacent comparisons in each strict ballot are needed in (4). Write voter i's order as

$$a_{i,1} \succ_{\pi_i} a_{i,2} \succ_{\pi_i} \cdots \succ_{\pi_i} a_{i,m},$$

and let

$$E_i = \{(a_{i,k}, a_{i,k+1}) : 1 \le k < m\}.$$

The adjacent inequalities imply all other ranking inequalities by transitivity.

We use the following standard finite-dimensional version of linear-programming duality.

Lemma 2 (Finite-dimensional strong LP duality)**.** *Consider a finite-dimensional linear program and its ordinary linear-programming dual. If the primal feasible region is nonempty and compact and the dual is feasible, then both programs attain optimal solutions and their optimal values are equal. No strict-feasibility or Slater-type hypothesis is required.*

For each $(a,b) \in E_i$, assign a nonnegative dual multiplier λ_{iab}. Define

$$h_i = \sum_{(a,b)\in E_i} \lambda_{iab}(e_a - e_b), \qquad H = \sum_{i=1}^{n} r_i h_i, \tag{8}$$

where e_a is the a-th standard basis vector of $\mathbb{R}^A$.

For $a \neq t$, let $\eta_a \geq 0$ be the multiplier for the constraint $S_t \geq S_a$, and let $y \in \mathbb{R}$ be the multiplier associated with $S_t + S_w = 1$. Set

$$c = y(e_w + e_t) - e_t - \sum_{a\neq t} \eta_a(e_t - e_a). \tag{9}$$

After substituting $S_a = \sum_i u_i(a)$, the ordinary dual of (5) is

$$\begin{aligned} &\text{minimize} && y, \\ &\text{subject to} && h_i + H \leq c && \text{for every } i, \\ &&& \lambda_{iab} \geq 0, \qquad \eta_a \geq 0, \end{aligned} \tag{10}$$

where vector inequalities are coordinatewise.

To verify both the dual signs and its weak-duality interpretation, take any primal-feasible and dual-feasible points. Then

$$\begin{aligned} 0 &\leq \sum_{i=1}^{n} h_i \cdot (u_i + r_i S) \\ &= \sum_{i=1}^{n} h_i \cdot u_i + \left(\sum_{i=1}^{n} r_i h_i\right) \cdot S \\ &= \sum_{i=1}^{n} (h_i + H) \cdot u_i \\ &\leq c \cdot S \\ &= y(S_w + S_t) - S_t - \sum_{a\neq t} \eta_a(S_t - S_a) \\ &= y - S_t - \sum_{a\neq t} \eta_a(S_t - S_a) \leq y - S_t. \end{aligned} \tag{11}$$

The first inequality follows because

$$h_i \cdot (u_i + r_i S) = \sum_{(a,b)\in E_i} \lambda_{iab}\big(u_i(a) + r_i S_a - u_i(b) - r_i S_b\big) \geq 0.$$

This also confirms directly that (10) supplies precisely the required coefficientwise upper bound.

The primal hypotheses of Lemma 2 were checked after (5). The dual is feasible: take

$$\lambda_{iab} = 0, \qquad \eta_a = 0, \qquad y = 1.$$

Then $h_i = H = 0$ and $c = e_w$, so $h_i + H \leq c$. Therefore strong duality applies, the dual optimum is attained, and its value is $q_t(r)$.

Lemma 3 (Dual sign lemma)**.** *Fix a finite vector $r \in [0,\infty)^n$, and take any optimal primal-dual pair for* (5) *and* (10)*. If $r_i > 0$, then*

$$h_i \cdot S \le 0. \tag{12}$$

Proof. At primal-dual optimality,

$$y = S_t = q_t(r).$$

Define

$$b = c - H. \tag{13}$$

Dual feasibility says

$$h_i \le b \qquad \text{for every } i. \tag{14}$$

We first establish

$$b_a \ge 0 \qquad \text{for every } a \ne t. \tag{15}$$

From (9),

$$c_a = \begin{cases} \eta_a, & a \notin \{t, w\}, \\ y + \eta_w, & a = w. \end{cases}$$

Since $y = q_t(r) \ge 1/2$, it follows that $c_a \ge 0$ whenever $a \ne t$.

Suppose, toward a contradiction, that $b_a < 0$ for some $a \ne t$. By (14),

$$h_i(a) \le b_a < 0 \qquad \text{for every } i.$$

Writing $R = \sum_i r_i$, we obtain

$$H_a = \sum_i r_i h_i(a) \le R b_a.$$

Since $c = b + H$,

$$c_a = b_a + H_a \le (1 + R) b_a < 0,$$

contradicting $c_a \ge 0$. This proves (15).

For any top prefix P of voter i's ranking,

$$h_i(P) := \sum_{a \in P} h_i(a) \ge 0. \tag{16}$$

Indeed, if $P = \{a_{i,1}, \ldots, a_{i,\ell}\}$ with $\ell < m$, then the telescoping definition (8) gives

$$h_i(P) = \lambda_{i, a_{i,\ell}, a_{i,\ell+1}} \ge 0.$$

For $P = A$, the same definition gives $h_i(A) = 0$. Combining (14) and (16), we have

$$b(P) \ge h_i(P) \ge 0 \qquad \text{for every top prefix } P \text{ of } \pi_i. \tag{17}$$

At optimality, the first and last expressions in (11) are both zero. Every intervening slack is nonnegative, so every inequality in that chain is an equality. In particular, for every voter i,

$$h_i \cdot (u_i + r_i S) = 0. \tag{18}$$

Moreover,

$$\begin{aligned}0 &= c \cdot S - \sum_i (h_i + H) \cdot u_i \\ &= \sum_{i=1}^{n} \sum_{a \in A} \big(c_a - H_a - h_i(a)\big) u_i(a) \\ &= \sum_{i=1}^{n} \sum_{a \in A} \big(b_a - h_i(a)\big) u_i(a).\end{aligned}$$

Every summand is nonnegative by (14) and $u_i(a) \geq 0$. Hence

$$\big(b_a - h_i(a)\big) u_i(a) = 0 \qquad \text{for every } i, a. \tag{19}$$

Summing (19) over a yields

$$h_i \cdot u_i = b \cdot u_i. \tag{20}$$

We claim that

$$b \cdot u_i \geq 0 \qquad \text{for every } i. \tag{21}$$

If $b_t \geq 0$, this follows immediately from (15) and $u_i(a) \geq 0$.

Suppose instead that $b_t < 0$, and let P_i be the top prefix of π_i ending at t. If $a \in P_i \setminus \{t\}$, then $a \succ_{\pi_i} t$. Chaining adjacent compatibility inequalities gives

$$u_i(a) + r_i S_a \geq u_i(t) + r_i S_t.$$

Because t is constrained to be welfare-maximizing, $S_t \geq S_a$, and therefore

$$u_i(a) \geq u_i(t) + r_i(S_t - S_a) \geq u_i(t). \tag{22}$$

All coefficients b_a outside the coordinate t are nonnegative. Consequently,

$$\begin{aligned}b \cdot u_i &\geq b_t u_i(t) + \sum_{a \in P_i \setminus \{t\}} b_a u_i(a) \\ &\geq u_i(t) \left(b_t + \sum_{a \in P_i \setminus \{t\}} b_a \right) \\ &= u_i(t) b(P_i) \geq 0,\end{aligned} \tag{23}$$

where the final inequality follows from (17). This proves (21) in all cases.

Finally, combining (18), (20), and (21), we obtain, whenever $r_i > 0$,

$$h_i \cdot S = -\frac{h_i \cdot u_i}{r_i} = -\frac{b \cdot u_i}{r_i} \leq 0.$$

□

6 Continuity and parametric monotonicity

We first prove continuity of q_t throughout the finite orthant, including at coordinates $r_i = 0$.

Set

$$L = 1 + \sum_{i=1}^{n} r_i, \qquad p_i = \frac{r_i}{L}, \qquad z_i(a) = \frac{u_i(a) + r_i S_a}{L}. \tag{24}$$

Then

$$\sum_i p_i < 1 \tag{25}$$

and

$$\sum_{i=1}^{n} z_i(a) = \frac{S_a + \left(\sum_i r_i\right) S_a}{L} = S_a. \tag{26}$$

Conversely,

$$u_i(a) = L\big(z_i(a) - p_i S_a\big). \tag{27}$$

Thus the feasible set of (5) is represented exactly by

$$\begin{aligned}
z_i(a) &\geq z_i(b) && \text{if } a \succ_{\pi_i} b, && (28)\\
z_i(a) &\geq p_i S_a && \text{for all } i, a, && (29)\\
S_a &= \sum_i z_i(a) && \text{for all } a, && (30)\\
S_t &\geq S_a && \text{for all } a, && (31)\\
S_t + S_w &= 1. && && (32)
\end{aligned}$$

Indeed, (29) is equivalent to nonnegativity of the utilities in (27). Furthermore,

$$\sum_i u_i(a) = L\left(S_a - \left(\sum_i p_i\right) S_a\right) = S_a,$$

because $\sum_i p_i = (L-1)/L$.

Let $\mathcal{K}(p)$ denote the polytope defined by (28)–(32). It is compact. To see this, use (27): all $u_i(a)$ are nonnegative and sum to S_a. Thus

$$0 \leq S_a \leq S_t \leq 1, \qquad 0 \leq u_i(a) \leq 1,$$

and

$$0 \leq z_i(a) = p_i S_a + \frac{u_i(a)}{L} \leq 2.$$

The defining constraints are closed.

Lemma 4 (Continuity at every finite parameter)**.** *The function $q_t(r)$ is continuous on $[0,\infty)^n$.*

Proof. The map

$$r \longmapsto p = \frac{r}{1 + \sum_j r_j}$$

is a continuous bijection between the finite nonnegative orthant and

$$\left\{ p \in [0,\infty)^n : \sum_i p_i < 1 \right\}.$$

It is therefore enough to prove continuity in p.

Fix such a vector p. By (25), one can choose numbers $s_i > p_i$ satisfying $\sum_i s_i = 1$; for example,

$$s_i = p_i + \frac{1 - \sum_j p_j}{n}.$$

Define

$$z_i^0(a) = \frac{s_i}{2} \qquad \text{for every } i, a. \tag{33}$$

Then $S_a^0 = 1/2$ for every alternative. Hence (28), (31), and (32) hold, while every parameter-dependent lower bound has the strict margin

$$z_i^0(a) - p_i S_a^0 = \frac{s_i - p_i}{2} > 0.$$

Let $p^{(k)} \to p$. For upper semicontinuity, take an optimizer $z^{(k)} \in \mathcal{K}(p^{(k)})$. The uniform bound $0 \le z_i^{(k)}(a) \le 2$ gives a convergent subsequence. Its limit satisfies all the constraints of $\mathcal{K}(p)$, since those constraints depend continuously on p. Therefore

$$\limsup_{k\to\infty} q_t(p^{(k)}) \le q_t(p).$$

For the reverse inequality, let z be an optimizer in $\mathcal{K}(p)$, and set

$$\varepsilon_k = \|p^{(k)} - p\|_\infty.$$

For all sufficiently large k, there is a constant $\delta > 0$, independent of k, such that

$$z_i^0(a) - p_i^{(k)} S_a^0 \ge \delta \qquad \text{for every } i, a.$$

Since $0 \le S_a \le 1$ and $z_i(a) \ge p_i S_a$,

$$z_i(a) - p_i^{(k)} S_a \ge -\varepsilon_k.$$

Choose

$$\theta_k = \frac{\varepsilon_k}{\varepsilon_k + \delta}$$

and define

$$\widehat{z}^{(k)} = (1 - \theta_k) z + \theta_k z^0.$$

All parameter-independent constraints are preserved by convexity. Moreover,

$$\begin{aligned} \widehat{z}_i^{(k)}(a) - p_i^{(k)} \widehat{S}_a^{(k)} &= (1-\theta_k)\big(z_i(a) - p_i^{(k)} S_a\big) + \theta_k\big(z_i^0(a) - p_i^{(k)} S_a^0\big) \\ &\ge -(1-\theta_k)\varepsilon_k + \theta_k \delta = 0. \end{aligned}$$

Thus $\widehat{z}^{(k)} \in \mathcal{K}(p^{(k)})$, and $\widehat{z}^{(k)} \to z$. It follows that

$$\liminf_{k\to\infty} q_t(p^{(k)}) \ge \lim_{k\to\infty} \widehat{S}_t^{(k)} = q_t(p).$$

This proves continuity. $\square$

We also use the following standard one-parameter LP envelope statement.

Lemma 5 (Parametric-LP envelope formula)**.** *Consider a finite linear program whose constraint-matrix entries depend affinely on a scalar parameter θ, while its objective coefficients are parameter-independent. Suppose a nonsingular primal-dual optimal basis remains valid on an open interval. Then the associated primal and dual basic solutions and the optimal value are differentiable on that interval. If the parameter-dependent inequalities are written as*

$$g_j(x, \theta) \leq 0$$

and have nonnegative multipliers λ_j under the Lagrangian convention

$$\mathcal{L}(x, \lambda; \theta) = c^\top x - \sum_j \lambda_j g_j(x, \theta),$$

then

$$\frac{dv}{d\theta} = -\sum_j \lambda_j \frac{\partial g_j}{\partial \theta}.$$

On every compact parameter interval, finitely many constraints imply that the interval can be partitioned into finitely many open basis-valid intervals and finitely many transition points. If the value function is continuous, a nonpositive derivative on every basis-valid interval implies that the value function is nonincreasing on the whole interval.

The finite-partition assertion follows from the usual basis description: for a fixed basis, the basic primal and dual solutions are rational functions of θ wherever the basis determinant is nonzero. Feasibility and optimality are determined by finitely many signs of rational functions. After clearing denominators, their changes can occur only at zeros of finitely many univariate polynomials. There are finitely many bases. Continuity then joins the basis-valid intervals and excludes isolated jumps or value spikes.

Proposition 1 (Finite-coordinate monotonicity)**.** *For every fixed profile π, winner w, and $t \in A \setminus \{w\}$, the function $q_t(r)$ is componentwise nonincreasing on $[0, \infty)^n$.*

Proof. Fix a coordinate i, hold all other coordinates fixed, and regard $r_i = \theta > 0$ as the scalar parameter. The only constraints whose coefficients depend on θ are voter i's order constraints, which in upper-bound form are

$$g_{iab}(U, \theta) = u_i(b) - u_i(a) + \theta(S_b - S_a) \leq 0, \qquad (a, b) \in E_i. \tag{34}$$

Their dependence on θ is affine.

On an interval where a nonsingular optimal basis remains valid, Lemma 5 gives

$$\begin{aligned} \frac{dq_t}{dr_i} &= -\sum_{(a,b)\in E_i} \lambda_{iab}(S_b - S_a) \\ &= \sum_{(a,b)\in E_i} \lambda_{iab}(S_a - S_b) = h_i \cdot S. \end{aligned} \tag{35}$$

The primal feasible region is nonempty and compact for every finite r, as checked after (5); the dual is feasible, as checked before Lemma 3; and the constraint dependence is affine in the scalar coordinate. Thus all hypotheses required for the basiswise envelope formula hold. Since $r_i > 0$, Lemma 3 gives

$$\frac{dq_t}{dr_i} = h_i \cdot S \leq 0$$

on every basis-valid interval.

There are finitely many bases, and Lemma 4 supplies continuity at all finite parameter values. Lemma 5 therefore shows that q_t is nonincreasing along every coordinate interval contained in $(0, \infty)$. Finally, for $x > 0$,

$$q_t(x) \le q_t(\varepsilon) \qquad (0 < \varepsilon < x).$$

Letting $\varepsilon \downarrow 0$ and using continuity proves $q_t(x) \le q_t(0)$. Hence monotonicity also includes the boundary $r_i = 0$. □

Since the map

$$\gamma_i \longmapsto r_i = \frac{\gamma_i}{n(1-\gamma_i)}$$

is increasing on $[0, 1)$, Proposition 1, Lemma 1, and monotonicity of $x/(1-x)$ yield

$$D_{\pi,w}(\gamma') \le D_{\pi,w}(\gamma) \tag{36}$$

whenever $\gamma' \ge \gamma$ and every coordinate of both vectors is strictly below 1.

7 Coordinates equal to one

It remains to pass rigorously to the full-public-spirit boundary. Fix $\gamma \in [0, 1]^n$, let

$$F = \{i : \gamma_i = 1\}, \qquad N = [n] \setminus F, \qquad k = |F|,$$

and suppose $k \ge 1$. For $L > 0$, define $\gamma^{(L)}$ by retaining the coordinates of voters in N and setting

$$\gamma_i^{(L)} = \frac{nL}{1+nL} \qquad (i \in F). \tag{37}$$

The corresponding scaled coefficient is exactly

$$\frac{\gamma_i^{(L)}}{n(1-\gamma_i^{(L)})} = L.$$

Lemma 6 (Exact endpoint convergence)**.** *For every fixed profile π and winner w,*

$$D_{\pi,w}(\gamma^{(L)}) \longrightarrow D_{\pi,w}(\gamma) \qquad \text{as } L \to \infty, \tag{38}$$

with convergence understood in the extended nonnegative reals.

Proof. We first prove the upper-limit inequality. By homogeneity, normalize every nonzero instance by

$$S_w + \max_{a \in A} S_a = 1. \tag{39}$$

Under this normalization,

$$0 \le u_i(a) \le S_a \le 1,$$

so all normalized instances lie in a fixed compact set.

Let $L_\ell \to \infty$, and let $U^{(\ell)}$ be normalized instances compatible with $(\pi, \gamma^{(L_\ell)})$. Every convergent subsequence has a limit U^*. The original, unscaled compatibility inequalities

$$(1-\gamma_i^{(L_\ell)})u_i^{(\ell)}(a) + \frac{\gamma_i^{(L_\ell)}}{n} S_a^{(\ell)} \ge (1-\gamma_i^{(L_\ell)})u_i^{(\ell)}(b) + \frac{\gamma_i^{(L_\ell)}}{n} S_b^{(\ell)}$$

depend continuously on both U and γ. Hence U^* is compatible with (π, γ).

If $D_{\pi,w}(\gamma) = +\infty$, the required upper-limit inequality is automatic. Suppose $D_{\pi,w}(\gamma) < \infty$. Then no normalized endpoint-compatible instance can have $S_w = 0$, because (39) would imply $\max_a S_a = 1$, producing infinite distortion. Compactness therefore implies that S_w has a positive minimum over the normalized endpoint-compatible set.

If the upper-limit inequality failed, one could choose normalized $U^{(\ell)}$ whose ratios remain strictly above $D_{\pi,w}(\gamma)$. A convergent subsequence would have an endpoint-compatible limit with positive winner welfare, and the ratios would converge to the ratio of that limit, contradicting the definition of $D_{\pi,w}(\gamma)$. Thus

$$\limsup_{L\to\infty} D_{\pi,w}(\gamma^{(L)}) \le D_{\pi,w}(\gamma). \tag{40}$$

For the reverse inequality, start from any nonzero endpoint-compatible utility matrix U, with welfare vector S. Define

$$A_a = \sum_{i\in F} u_i(a) = S_a - \sum_{j\in N} u_j(a). \tag{41}$$

At the endpoint, every voter in F ranks alternatives solely according to S. We may therefore redistribute the full voters' utilities so that each of them initially receives A_a/k from alternative a. This preserves all welfare values and every voter's compatibility.

For $j \in N$, set

$$r_j = \frac{\gamma_j}{n(1-\gamma_j)}, \qquad R_N = \sum_{j\in N} r_j.$$

Let

$$\varepsilon = \frac{1}{L}, \qquad d = k + \varepsilon(1+R_N), \qquad \Delta S_a = -\frac{\varepsilon A_a}{d}. \tag{42}$$

Define

$$S_a^{(L)} = S_a + \Delta S_a, \tag{43}$$

and utilities

$$u_j^{(L)}(a) = u_j(a) - r_j \Delta S_a \quad (j\in N), \qquad u_i^{(L)}(a) = \frac{A_a}{d} \quad (i \in F). \tag{44}$$

Because $A_a \ge 0$ and $\Delta S_a \le 0$, all these utilities are nonnegative.

Their column sums are exactly the prescribed welfare values. Indeed,

$$\begin{aligned}
\sum_{i=1}^{n} u_i^{(L)}(a) &= S_a - A_a - R_N \Delta S_a + \frac{kA_a}{d} \\
&= S_a - A_a + \frac{\varepsilon R_N A_a}{d} + \frac{kA_a}{d} \\
&= S_a + \frac{-d + \varepsilon R_N + k}{d} A_a \\
&= S_a - \frac{\varepsilon A_a}{d} = S_a^{(L)}.
\end{aligned}$$

For every $j \in N$, the scaled score is unchanged:

$$\begin{aligned}
u_j^{(L)}(a) + r_j S_a^{(L)} &= u_j(a) - r_j\Delta S_a + r_j(S_a + \Delta S_a) \\
&= u_j(a) + r_j S_a.
\end{aligned} \tag{45}$$

Thus voter j's entire weak score ordering, including every tie, is preserved.

For every $i \in F$, the scaled score under $\gamma_i^{(L)}$ is

$$\begin{aligned} u_i^{(L)}(a) + LS_a^{(L)} &= \frac{A_a}{d} + L\left(S_a - \frac{\varepsilon A_a}{d}\right) \\ &= LS_a, \end{aligned} \tag{46}$$

because $L\varepsilon = 1$. Endpoint compatibility for a full voter says

$$a \succ_{\pi_i} b \quad \Longrightarrow \quad S_a \geq S_b.$$

Equation (46) therefore makes the same strict ballot compatible at every finite L. Hence $U^{(L)}$ is compatible with $(\pi, \gamma^{(L)})$.

Moreover,

$$S_a^{(L)} \longrightarrow S_a \qquad \text{for every } a.$$

If $S_w > 0$, the corresponding welfare ratios converge:

$$\frac{\max_a S_a^{(L)}}{S_w^{(L)}} \longrightarrow \frac{\max_a S_a}{S_w}.$$

If $S_w = 0$, nonnegativity implies $A_w = 0$, and (42) gives

$$S_w^{(L)} = S_w = 0$$

exactly. Since $U \neq 0$, $\max_a S_a > 0$, and $\max_a S_a^{(L)} > 0$ for all sufficiently large L. Thus an infinite endpoint ratio remains infinite in the approximating instances.

If $D_{\pi,w}(\gamma) < \infty$, compact normalization gives an endpoint optimizer, and the construction approximates its ratio. If $D_{\pi,w}(\gamma) = +\infty$, compactness of the normalized feasible set produces an endpoint-compatible instance with $S_w = 0$, and the construction preserves its infinite ratio. Therefore

$$\liminf_{L\to\infty} D_{\pi,w}(\gamma^{(L)}) \geq D_{\pi,w}(\gamma). \tag{47}$$

Together with (40), this proves (38). $\square$

The endpoint argument is stated directly for $D_{\pi,w}$, rather than for a fixed face q_t. The construction may change which alternative among welfare ties is selected as a welfare maximizer, but this is immaterial because the numerator of $D_{\pi,w}$ is $\max_a S_a$.

8 Completion of the proof

Proof of Theorem 1. The case $m = 1$ was settled at the outset, so suppose $m \geq 2$. Fix a profile π and put $w = f(\pi)$.

For a vector $\alpha \in [0,1]^n$, define its finite approximation by

$$\alpha_i^{(L)} = \begin{cases} \alpha_i, & \alpha_i < 1, \\ \dfrac{nL}{1+nL}, & \alpha_i = 1. \end{cases}$$

If $\gamma \leq \gamma'$, then for all sufficiently large L,

$$\gamma^{(L)} \leq (\gamma')^{(L)} \tag{48}$$

componentwise. Indeed:

- if both coordinates are below 1, their original order is unchanged;
- if both are equal to 1, their approximations are equal;
- if $\gamma_i < 1 = \gamma'_i$, then $\frac{nL}{1+nL} \uparrow 1$, so it eventually exceeds γ_i;
- the case $\gamma_i = 1 > \gamma'_i$ is excluded by $\gamma_i \leq \gamma'_i$.

Every coordinate of the approximating vectors is strictly below 1. Therefore the finite-coordinate result (36) gives

$$D_{\pi,w}\big((\gamma')^{(L)}\big) \leq D_{\pi,w}\big(\gamma^{(L)}\big)$$

for all sufficiently large L. Applying Lemma 6 separately to γ and γ' and letting $L \to \infty$ yields

$$D_{\pi,w}(\gamma') \leq D_{\pi,w}(\gamma).$$

This holds for every strict profile π. Consequently,

$$\begin{aligned}
\operatorname{dist}(f, \gamma') &= \max_{\pi \in \mathcal{R}(A)^n} D_{\pi, f(\pi)}(\gamma') \\
&\leq \max_{\pi \in \mathcal{R}(A)^n} D_{\pi, f(\pi)}(\gamma) = \operatorname{dist}(f, \gamma),
\end{aligned}$$

where (1) was used in the first and last equalities. □

The theorem covers every deterministic rule, every positive number of voters, the entire public-spirit cube $[0,1]^n$, and every finite number of alternatives. In particular, it resolves the target case $m \geq 4$ completely. There is no remaining mathematical gap or unproved parameter regime.

References

[1] Bailey Flanigan, Ariel D. Procaccia, Sven Wang. *Distortion Under Public-Spirited Voting.* Economics and Computation (EC), 2023. https://doi.org/10.1145/3580507.3597722